

Mofe Omolade

(825) 561-5864 | omolade.mofe@gmail.com | linkedin.com/in/momolade | mofeomolade.github.io

EDUCATION

University of British Columbia, *BASc in Computer Engineering*

Expected 2029

Relevant Coursework: Circuit Analysis, Electricity & Magnetism, Linear Algebra, Embedded C, Engineering Design.

SKILLS

- Hardware & Protocols: PCBA Design, Power Electronics, CAN, SPI, I2C, UART, RS-485, STM32, ESP32, Arduino.
- Tools: Altium, KiCAD, LTspice, MATLAB, Git, VS Code, STM32 CubeIDE.
- Languages: C/C++, Python.
- Lab: Oscilloscopes, Logic Analyzers, DMM, Function Generators, Soldering, Circuit Debugging.

WORK EXPERIENCE

FulcrumAir, *Electronics Engineering Intern*, Calgary, AB

May 2026 - Present

- Designed power distribution, communications breakout, and motor control PCBs for industrial robots in Altium.
- Designed an 8-layer motor control PCB in Altium around Teensy 4.1, IFX007T, MAX33040E, and LMR50410-Q1.
- Developed firmware for motor control, CAN communication, and linear actuator positioning feedback.
- Implemented a power hold-up system to trigger EEPROM data write within <5ms upon detecting voltage disconnect.

University of Calgary, *Electrical Engineering Research Assistant*, Calgary, AB

July 2023 - August 2023

- Modelled and simulated Op-Amp oscillator circuits in LTspice to support research in amplifier-based oscillators.
- Contributed to academic projects across faculties and presented findings at symposiums and research boards.

DESIGN TEAM

UBC Rover, *Electrical Subteam*, Vancouver, BC

September 2025 - Present

- Collaborated with chassis and software subteams to design a test rover for autonomous navigation training.
- Developed power architecture, integrated UART serial communication, and wrote low-level embedded firmware.
- Designed a powertrain prototype around the L298N full-bridge motor driver and Nucleo F446RE MCU.
- Implemented C firmware using STM32 HAL for command parsing, motor feedback, and UART communication.

PROJECTS

IMU Pitch, Roll & Yaw Calculator

- Developed low-level C++ drivers for ESP32 to parse MPU-6050 IMU accelerometer and gyroscope registers via I2C.
- Applied a sensor-fusion Euler complementary filter and a deadband filter to account for angular integration drift.
- Applied trigonometry to accelerometer gravity vectors and fused with gyroscope readings to calculate pitch and roll.

Speedometer Heads-Up Display

- Designed a low-latency heads-up display using an Arduino Nano, GY-NEO6MV2 module, and SPI OLED display.
- Wrote C++ firmware to parse NMEA data streams to compute velocity within 5% of the factory speedometer.
- Safely integrated with vehicle 14V input using a USB-C to DIN converter to supply the Arduino Vin.

Synchronous 24V-5V 2A Buck Converter

- Designed buck topology with dual N-channel MOSFETs, floating gate drives, and analog Type II loop compensation.
- Validated open-loop dynamics, small-signal Bode stability, and transient response in LTspice simulations.
- Performed 4-layer PCB layout in Altium, minimizing ground and switching loops to maintain signal integrity.